Internship Proposal: Development of a Custom Interaction and Control System for an Android-Based Service Robot

HESTIM Engineering & Business School

Overview

This internship focuses on the development of a custom interaction and control system for a service robot operating on an Android platform. The robot currently depends on a proprietary application that limits accessibility and flexibility. The objective is to design and implement an independent and extensible solution that enables full interaction and control without relying on the original application.

Problem Context

The robot combines an Android-based interface with an internal control system responsible for motion and actuation. However, the absence of access to the original application and lack of official documentation introduce significant challenges. The main task is to identify and reconstruct the communication mechanism between the user interface and the robot's control system, while addressing network constraints and unknown protocols.

Project Objectives

The main objectives of this internship are:

- Establish communication between the robot and an external device
- Identify communication endpoints (IP, ports, protocols)
- Capture and analyze network traffic to understand the protocol
- Decode and reconstruct control commands for robot movement
- Develop a custom control interface (web, Python, or Android)
- Implement basic human-robot interaction features (UI and voice)
- Integrate all components into a unified system

Expected Deliverables

At the end of the internship, students are expected to provide:

- A working communication interface with the robot
- A functional control system for robot movement
- A user interface (web or Android application)
- A basic interaction module (touch and/or voice)
- A technical report including system architecture and protocol analysis

Skills & Technologies

Students will develop and apply skills in:

- Computer networking (TCP/IP, sockets, WiFi)
- Protocol analysis and reverse engineering
- Python programming
- Web development (HTML, JavaScript)
- Android application development (optional)
- Human-Robot Interaction (HRI)

Timeline

The internship is planned over a duration of three to four months. The initial phase focuses on understanding the system architecture and establishing network connectivity with the robot. This is followed by a protocol exploration phase, during which students analyze communication patterns and identify control commands.In the next phase, students will design and develop the control interface and interaction modules. The final phase is dedicated to system integration, testing, and validation, culminating in a functional demonstration and complete documentation of the work.

Candidate Profile

This internship is intended for:

- 3rd or 4th year engineering students
- Background in computer science, robotics, embedded systems, or similar fields
- Basic knowledge of programming and networking
- Interest in robotics

Optional Extensions

Depending on progress, students may explore:

- Autonomous behaviors
- Vision-based interaction using onboard camera
- Chatbot integration
- Multi-robot coordination

Application & Contact Information

Students interested in this internship are invited to submit their application, including a **CV** and a brief **motivation letter**, outlining their background and interest in the project.

For further information or to apply, please contact:

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